06 / Decision – full 1-3-9-27 screen

What bounded conclusion or next action does the evidence justify?

EX02 / UMI | Source analysis | C1 | P: paper; W: project; R: repository

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Source analysis | C1: On reported dynamic tossing, UMI latency matching improves object-level success with the same trained policy.

Generate grounded experience

G1 / Grounded physical problem

G1.1 Mechanism: Is the claimed mechanism explicit, physically credible and open to challenge?

G1.2 Closed loop: Are the implemented body, observations, controller, actions, environment and human roles identified?

G1.3 Task boundary: Are task conditions, episode boundaries, outcomes and exclusions stated?

G2 / Measurable embodiment

G2.1 Embodiment: Is the embodiment or physical coupling realized and identified at the claimed stage?

G2.2 Observability: Are measurements and references suitable for the claimed outcomes, with relevant validity limits?

G2.3 Interfaces: Are consequential interfaces, action meanings, units and timing described or verified?

G3 / Controlled physical evidence

G3.1 Commissioning: What limits, resets, interventions, fault handling and recovery are described or tested?

G3.2 Experience record: What observations, actions, outcomes and contextual records are available or described?

G3.3 Grounding gap: How are model, measurement or interpretation connected to physical evidence under comparable conditions?

Use experience for claims

U1 / Bounded claim logic

U1.1 Claim: Can the available evidence support or challenge the specific claim, without expanding its scope?

U1.2 Role of experience: Is the role of experience in learning, inference, adaptation, evaluation or design insight explicit?

U1.3 Comparators: Are comparators or alternative explanations appropriate to the contribution being claimed?

U2 / Representative experience

U2.1 Correspondence: Are data, model, policy, embodiment and evaluation configurations linked where needed?

U2.2 Coverage: Are coverage, sampling, failures, evaluation partitions and leakage addressed for the claim?

U2.3 Governance: What provenance, rights, human participation and data safeguards are reported where relevant?

U3 / Reproducibility and challenge

U3.1 Reconstruction: What is available to reconstruct the system and evaluation, and what remains unverified?

U3.2 Evaluation: Are protocol, denominators, repetitions, uncertainty and exclusions reported appropriately?

U3.3 Failure analysis: Are failure modes, interventions, competing explanations and negative results examined?

Manage evidence through lifecycle

M1 / Evidence traceability

M1.1 Identification: Are claims, configurations, datasets and experiments distinguishable and source-linked?

M1.2 Change control: Are variants, updates or corrections traceable, and is their effect on conclusions assessed?

M1.3 Counterevidence: Are material failures, null findings, ambiguous outcomes and counterevidence retained?

M2 / Resources, burdens and benefits

M2.1 Resources: Are relevant hardware, compute, time, skills and maintenance resources stated?

M2.2 Data burden: Are collection, annotation, reset, storage and human effort addressed where consequential?

M2.3 Benefit: Is the claimed benefit justified against costs and relevant alternatives within the study?

M3 / Continuing operation

M3.1 Monitoring: What detects mismatch, degradation, invalid measurements or faults in the claimed setting?

M3.2 Operations: What human authority, maintenance, recovery and update practices are described for this stage?

M3.3 Sustainment: What duration, units, contexts or repeated-use evidence supports the claimed persistence?

Support (strength for the bounded question)

0 contradicted; 1 weak; 2 partial; 3 adequate within scope; 4 convergent. NR not reported in inspected scope; NE not examined; NA not applicable with a reason.

Evidence context (separate from origin, support and access)

E1 reasoning/design/plan; E2 executed analysis/simulation/digital artifact; E3 physical/integrated experiments, including externally reported results; E4 specified physical coverage/auditability; E5 independent reproduction/sustained operation within scope. Not a universal quality ladder. NR/NE/NA: context normally a dash.

06 / Decision – evidence and action record

Source analysis: assess the selected claim; retain unexamined questions.

EX02 / UMI | Source analysis | C1 | P: paper; W: project; R: repository

C1 / UMI / Selected source claim; physical/simulated origin retained; no independent reproduction

G1 / Grounded physical problem

Check	Evidence ID + basis	Support	Context
G1.1	P V-B tossing design	3	E1
G1.2	P Fig5 collection/robot	2	E3
G1.3	P AppA-C bin outcome	3	E1

State: Scoped / Gap: local reproduction NE
Analyst: Check named task/outcome.

G2 / Measurable embodiment

Check	Evidence ID + basis	Support	Context
G2.1	P Hardware gripper	2	E3
G2.2	P AppA-C manual labels	2	E3
G2.3	P Fig5 latency semantics	2	E1

State: Scoped / Gap: raw interface audit NE
Analyst: Audit units/order offline.

G3 / Controlled physical evidence

Check	Evidence ID + basis	Support	Context
G3.1	NE local stop/limit test	NE	-
G3.2	P V-B 280 demos	2	E3
G3.3	P V-B 105/120 vs69/120	3	E3

State: Open / Gap: local execution NE
Analyst: Inspect failures and timing.

U1 / Bounded claim logic

Check	Evidence ID + basis	Support	Context
U1.1	P V-B bounded effect	3	E3
U1.2	P Policy imitation role	3	E1
U1.3	P V-B same-policy test	3	E3

State: Scoped / Gap: scope kept bounded
Analyst: Match comparator and scope.

U2 / Representative experience

Check	Evidence ID + basis	Support	Context
U2.1	R session route only	2	E1
U2.2	P V-B episode/object units	2	E1
U2.3	NE rights review	NE	-

State: Open / Gap: split/rights audit NE
Analyst: Pin origin/split/access.

U3 / Reproducibility and challenge

Check	Evidence ID + basis	Support	Context
U3.1	R pipeline not executed	2	E1
U3.2	P V-B object counts	3	E3
U3.3	P V-B release failures	3	E3

State: Scoped / Gap: no independent rerun
Analyst: Verify result and trial unit.

M1 / Evidence traceability

Check	Evidence ID + basis	Support	Context
M1.1	P v3 V-B comparison	3	E1
M1.2	NE code release pin	NE	-
M1.3	R reconstruction issues	2	E1

State: Open / Gap: release change audit NE
Analyst: Freeze source/config IDs.

M2 / Resources, burdens and benefits

Check	Evidence ID + basis	Support	Context
M2.1	P Hardware build	2	E1
M2.2	P V-B demo burden	2	E3
M2.3	P V-B latency benefit	2	E3

State: Scoped / Gap: full burden incomplete
Analyst: Separate counted burdens.

M3 / Continuing operation

Check	Evidence ID + basis	Support	Context
M3.1	NE local fault audit	NE	-
M3.2	P AppA-C supervision	2	E1
M3.3	NE long-run evidence	NE	-

State: Open / Gap: long-run evidence NE
Analyst: Require a scoped next test.

Choose the conclusion for the declared mode

Source analysis: support is claim-specific, not a course grade. E1 description/plan; E2 reported simulation/digital experiment; E3 reported physical experiment; NE not examined. No score average or deployment approval.

Project-development action: not assessed here. A published result does not authorize the student team to deploy a robot.

Bounded conclusion: In the reported tossing comparison, matching achieves 105/120 object successes versus 69/120 without it. This supports the timing interface, not arbitrary-platform transfer.

Follow-up: Analyst: verify V-B counts and App A clocks/outcome rule; inspect a delayed-release failure; do not deploy from these notes alone.

Reassess if data, calibration, implementation or source conclusions change. Next project action requires its own evidence.

Worksheet 6 / Decision - evidence and team review

EX02 / UMI | SOURCE ANALYSIS EXAMPLE | Original review-page layout

Example ID: EX02 Project: UMI Claim ID: C1 Revision: draft 2

Mode: Source analysis Pages: canvas + this review; Decision also includes both A3 sides

Evidence scope: selected paper text/tables checked 20 Oct 2026; no raw dataset download or independent experiment.

Evidence ID	Artifact/source + location (clickable)	What it supports; limitation / access gap
P	arxiv.org/html/2402.10329v3	Paper: method/results; reported evidence, not our measurement.
W	real.stanford.edu/umi	Project: system/videos; selected examples do not establish reliability.
R	github.com/real-stanford/universal_manipulation_interface	Repository: linked route; code, raw data and current commit not audited.

Next action / owner / verification:

Analyst: inspect one delayed-release failure and one accepted session; resolve clock/SLAM gaps before a reproduction.

Teaching example: no actual student team or signatures. Source analysis is not a reproduced result; blank signature fields are intentional.

1. Name / student ID: _____

Signature: _____ Date: _____

2. Name / student ID: _____

Signature: _____ Date: _____

3. Name / student ID: _____

Signature: _____ Date: _____

4. Name / student ID: _____

Signature: _____ Date: _____

5. Name / student ID: _____

Signature: _____ Date: _____

6. Name / student ID: _____

Signature: _____ Date: _____

7. Name / student ID: _____

Signature: _____ Date: _____

8. Name / student ID: _____

Signature: _____ Date: _____

P/W/R identify linked public sources above. NE: not examined in this brief example, not evidence of failure. Support/context labels are provisional, not grades.